Compressing LLMs using Distillation

Adwait Agashe . Muskan Beriwal . Prekshi Vyas . Uday Malhotra

Introduction

Knowledge Distillation

Teacher Model → Model Distillation → Student Model

Gemini

High Performance

- Expensive
- Overkill for simple tasks

Student Model:

- Off the shelf smaller model
- Pruned version of the teacher
- Blank model trained from scratch

Easier and Cheaper to train and deploy

Helps protect proprietary architecture

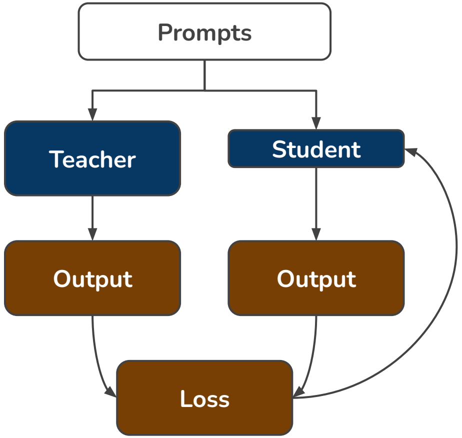
Stages:

1. Knowledge Elicitation

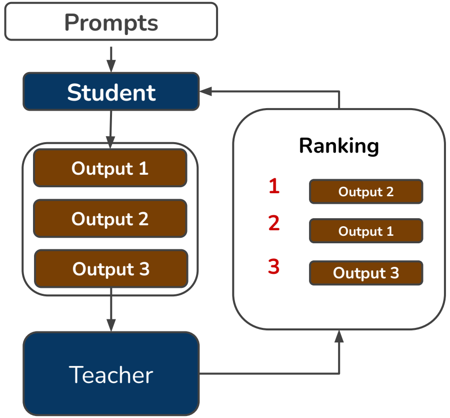
Use the teacher model on an unlabelled dataset to generate task labels

2. Distillation Algorithm

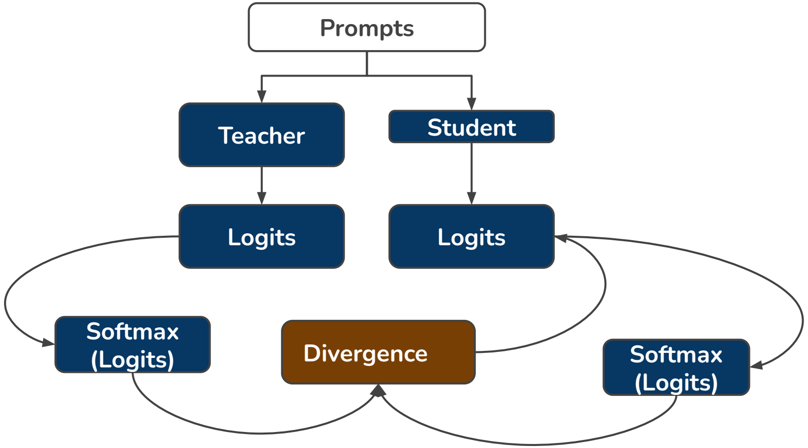
Distillation Algorithms



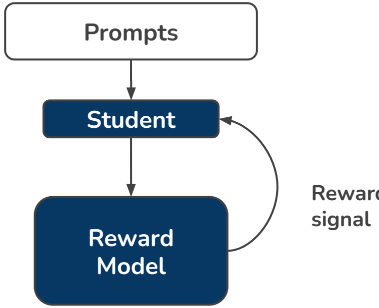
Supervised Fine Tuning



Ranking Based



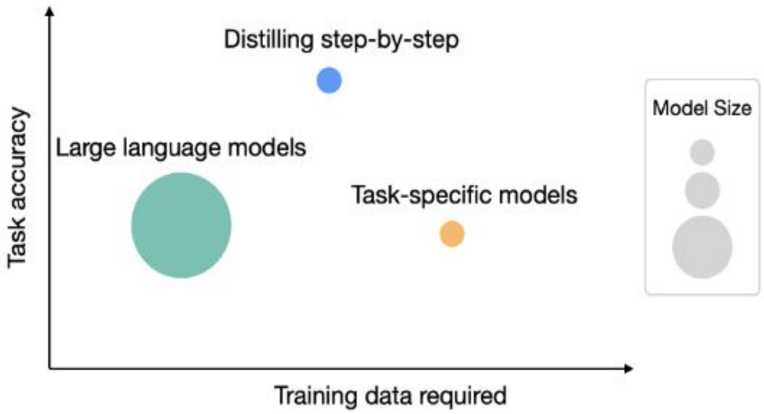
Divergence and Similarity



Reinforcement Learning

Paper 1: Distilling Step-by-Step! (ACL- 2023)

Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes



Motivation:

- **LLMs**: Resource-intensive and impractical for real-world deployment.
- **Smaller models**: Feasible but require extensive labeled data & struggle with performance.
- **Solution**: Train task-specific models using LLM-generated rationales for efficient learning with less data and compute.

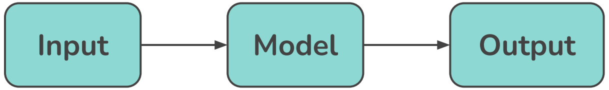
Background:

- **Traditional solutions**:
  - **Fine-tuning**: Smaller models need large labeled datasets.
  - **Knowledge distillation**: Requires vast unlabeled data but sacrifices reasoning quality.
- **Opportunity**: LLMs can generate rationales (step-by-step reasoning) that guide smaller models & improve performance with fewer resources.

Intuition

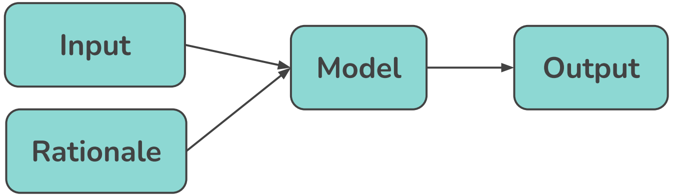
Standard Fine Tuning and Distillation:

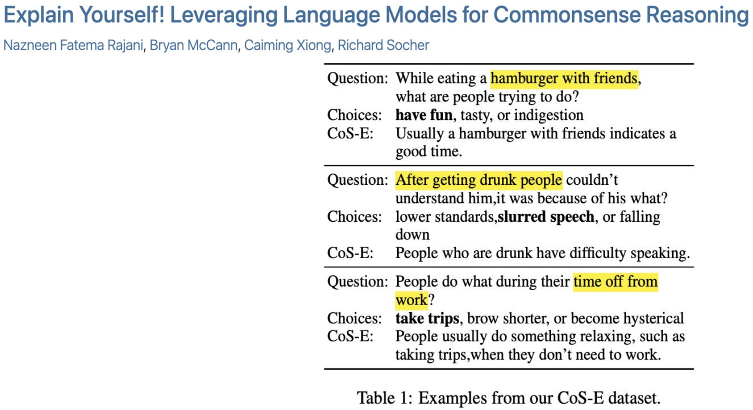
- ŷ can be human labels (standard fine-tuning) or LLM-generated labels (standard distillation)



Distillation using Rationales - Approach 1:

- Treat **rationales** as extra model **input**





Explain Yourself! Leveraging Language Models for Commonsense Reasoning

Nazneen Fatema Rajani, Bryan McCann, Caiming Xiong, Richard Socher

Question:	While eating a hamburger with friends, what are people trying to do?
Choices:	**have fun**, tasty, or indigestion
CoS-E:	Usually a hamburger with friends indicates a good time.
Question:	After getting drunk people couldn't understand him,it was because of his what?
Choices:	lower standards,**slurred speech**, or falling down
CoS-E:	People who are drunk have difficulty speaking.
Question:	People do what during their time off from work?
Choices:	**take trips**, brow shorter, or become hysterical
CoS-E:	People usually do something relaxing, such as taking trips,when they don't need to work.

Table 1: Examples from our CoS-E dataset.

PINTO: Faithful Language Reasoning Using Prompt-Generated Rationales

PeiFeng Wang, Aaron Chan, Filip Ilievski, Muhao Chen, Xiang Ren

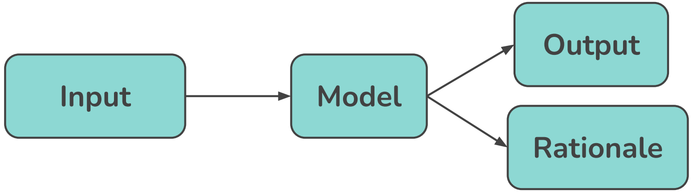
Published: 01 Feb 2023, Last Modified: 25 Nov 2024 ICLR 2023 poster Readers: Everyone Show Bibtex Show Revisions

Keywords: Commonsense reasoning, free-text rationale, rationale generation, faithful reasoning

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \ell(f(x_i, \hat{r}_i), \hat{y}_i)$$

Distillation using Rationales - Approach 2:

- Treat **rationales as supervision** rather than inputs.
- Using rationales to improve the training process, enabling smaller models to generalize better with less data

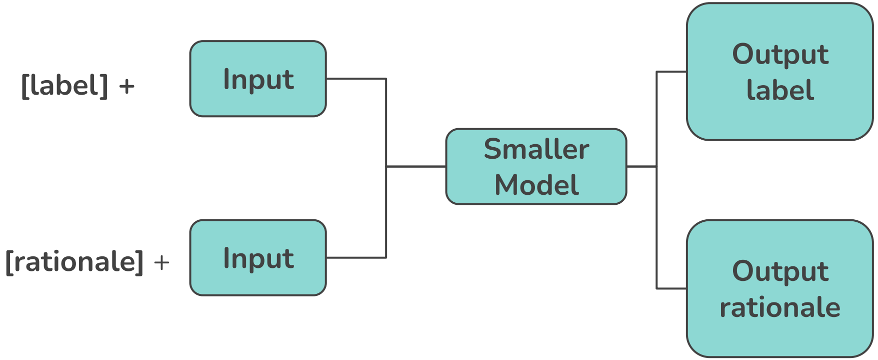


$$\mathcal{L}_{\text{label}} = \frac{1}{N}\sum_{i=1}^{N}\ell(f(x_i), \hat{y}_i)$$

$$\mathcal{L}_{\text{rationale}} = \frac{1}{N}\sum_{i=1}^{N}\ell(f(x_i), \hat{r}_i)$$

$$\mathcal{L} = \mathcal{L}_{\text{label}} + \lambda\mathcal{L}_{\text{rationale}}$$

Training with *task-prefixes*:



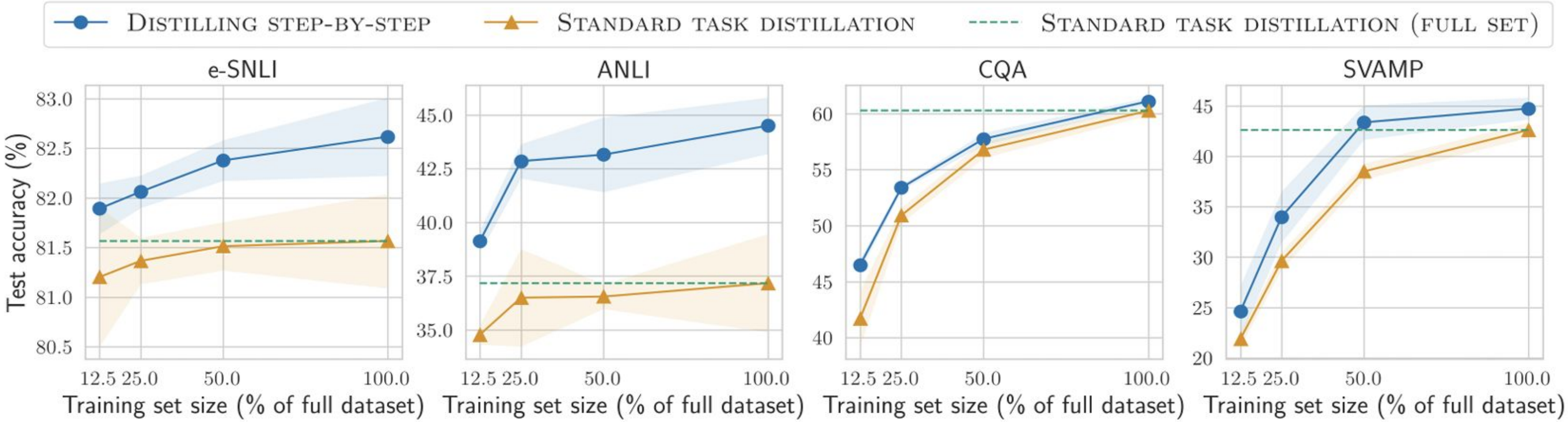
Experiments & Results

1. Reduced Training Data:

e-SNLI and ANLI → Natural Language Inference
CQA → Common Sense Reasoning
SVAMP → Math Word Problems

Orange → Standard Distillation
Blue → Distil Step By Step
Green → Full Dataset Standard Distillation

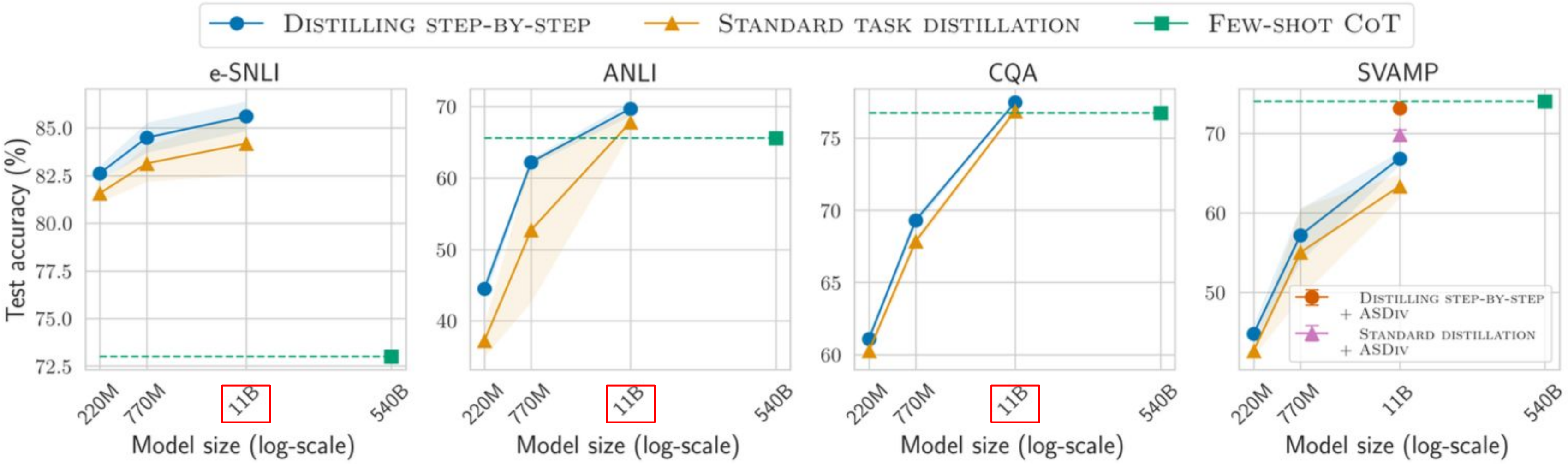
1. Outperforms Standard Distillation for all Benchmarks
2. 12.5% Data → Better than Full Distill for e-SNLI and ANLI
3. Huge reduction in training data required while maintaining performance



2. Reduced Model Size:

Orange → Standard Distillation
Blue → Distil Step By Step
Green → Few Shot CoT for 540B Model

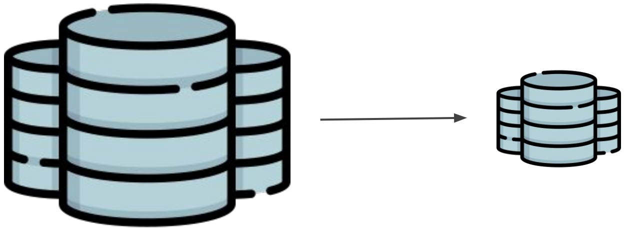
1. Big improvements in accuracy for smaller models
2. 11B Model + Distill Step by Step better than 540B + Few Shot CoT
3. Can use 50x smaller model to get similar performance



Real World Impact

Data Efficiency:

- Significant reduction in labeled and unlabeled data required.
- Similar accuracy with 12.5% of original training data.
- Reduction in training compute required for fine tuning model.



Deployment Feasibility:

- Deploy 50x - 700x smaller models without substantial drop in performance.
- Significant improvement in inference cost and latency.

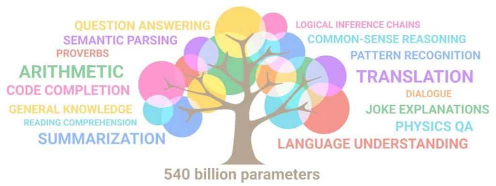
Demo Experiments

Common Sense QA Task

There are 10 apples on an apple tree. Three fall off. Now there are X apples. What is this an example of?

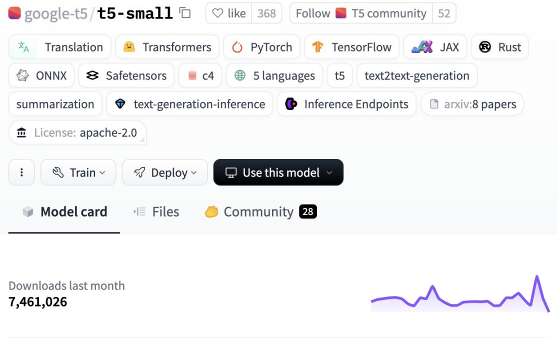
Answer Choices:
(a) park
(b) coloring book
(c) garden center
(d) **math problem**
(e) gravity

Larger Model (PaLM)



- 540B Parameters
- Used to generate labels and rationales
- Labels and Rationales provided by original authors.

Smaller Model

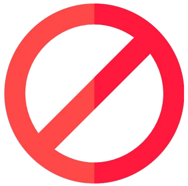


- 60M Parameters
- Used as student model for distillation

Demo Results

No Distillation

- CQA Test set run directly on T5 Small
- 0% Accuracy
- Model produces random completions without predicting answers

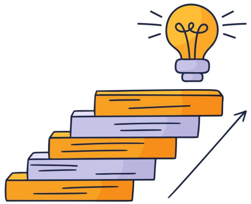


Standard Distillation

- Trained for Label Prediction using PaLM labels
- 14.08% Accuracy after 5 Epochs

Distill Step By Step

- Trained for Label + Rationale Prediction using PaLM predictions
- 25.47% Accuracy after 5 epochs.



Paper 2: MiniLLM: Knowledge Distillation of Large Language Models (ICLR - 2024)

Background:

- Knowledge Distillation (KD):
  - Size benefits with increase in larger and more powerful models
  - White Box KD vs Black Box KD
    - Scope of White box KD not explored
    - Important for research and can help improve performance

- Standard KD Objective
  - **Forward Kullback Leibler (KL) Divergence**

$$\mathrm{KL}[p||q_\theta] = \mathbb{E}_{\boldsymbol{x}\sim p_{\boldsymbol{x}}, \boldsymbol{y}\sim p'} \log \frac{p(\boldsymbol{y}|\boldsymbol{x})}{q_\theta(\boldsymbol{y}|\boldsymbol{x})}$$

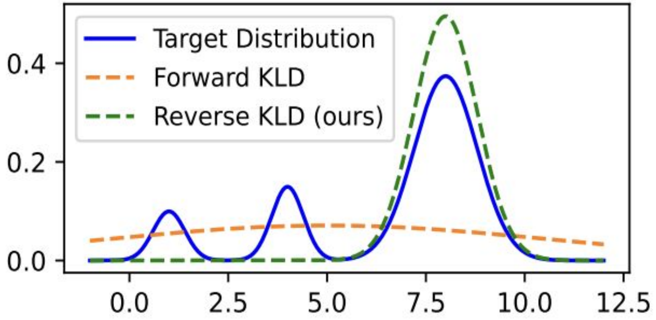
Motivation

- Forward KL Divergence is good for closed text generation
  - Limited classes
  - Harder for open text generation
    - Teacher’s capacity vs Student’s capacity

- **Fix: Reverse KL Divergence (inspired from CV & RL)**

$$\arg\min_{\theta}\left[-\mathop{\mathbb{E}}_{\boldsymbol{x}\sim p_{\boldsymbol{x}},\boldsymbol{y}\sim q_{\theta}}\log\frac{p(\boldsymbol{y}|\boldsymbol{x})}{q_{\theta}(\boldsymbol{y}|\boldsymbol{x})}\right]$$

- Intuition:
  - Reverse KL Divergence optimizes higher modes



Algorithm

- Use **Policy Gradient Theorem:**

$$\nabla \mathcal{L}(\theta) = - \mathop{\mathbb{E}}_{\boldsymbol{x} \sim p_{\boldsymbol{x}}, \boldsymbol{y} \sim q_\theta(\cdot|\boldsymbol{x})} \sum_{t=1}^{T} (R_t - 1) \nabla \log q_\theta(y_t | \boldsymbol{y}_{<t}, \boldsymbol{x})$$

- R(t) is *summed* forward KL Divergence:

$$R_t = \sum_{t'=t}^{T} \log \frac{p(y_{t'} | \boldsymbol{y}_{<t'}, \boldsymbol{x})}{q_\theta(y_{t'} | \boldsymbol{y}_{<t'}, \boldsymbol{x})}$$

  - Issues:
    - R(t) favors shorter sentences. **Fix: Length Normalization**
    - R(t) gives higher score to degenerate sentences. **Fix: Teacher-Mixed Sampling**
    - Errors at each timestep accumulate. **Fix: Single Step Decomposition**

Experimental Setup

- **Base (Teacher) Models:** GPT-2, OPT, LLaMA (varied sizes for teacher and student)
- **Evaluation Metrics:**
  - Rouge-L: instruction evaluation
  - ChatGPT-4: numerical scale
  - Human Evaluation: nominal scale
- **Baselines:**
  - SFT w/o KD: direct fine tuning
  - KD: regular KD with teacher supervision
  - SeqKD: KD with teacher responses for fine tuning

Major Results & Impact

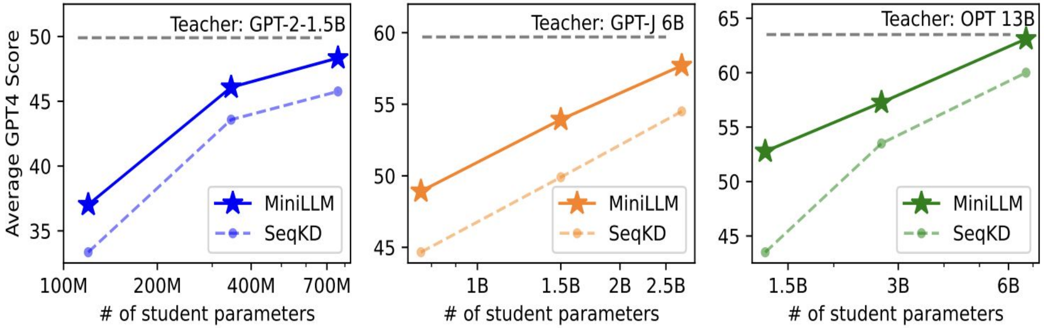
1. Outperforms baselines

2. High overlap with ground truth (Rouge-L)

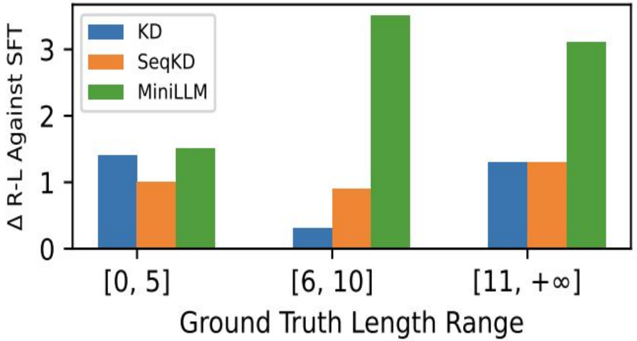
3. Produces more diverse responses

Model	#Params	Method	DollyEval		SelfInst		VicunaEval		S-NI	UnNI
			GPT4	R-L	GPT4	R-L	GPT4	R-L	R-L	R-L
GPT-2	1.5B	Teacher	58.4	27.6	42.9	14.3	48.6	16.3	27.6	31.8
	120M	SFT w/o KD	38.6	23.3	26.3	10.0	32.8	14.7	16.3	18.5
		KD	40.3	22.8	27.8	10.8	31.9	13.4	19.7	22.0
		SeqKD	41.2	22.7	26.2	10.1	31.0	14.3	16.4	18.8
		MiniLLM	**44.7**	**24.6**	**29.2**	**13.2**	**34.1**	**16.9***	**25.3**	**26.6**
	340M	SFT w/o KD	51.9	**25.5**	39.6	13.0	42.3	16.0	25.1	32.0
		KD	51.6	25.0	39.2	12.0	42.8	15.4	23.7	31.0
		SeqKD	50.5	25.3	39.0	12.6	**43.0**	16.9*	22.9	30.2
		MiniLLM	**52.2**	25.4	**40.5**	**15.6**	42.6	**17.7***	**27.4**	**34.5**
	760M	SFT w/o KD	50.7	25.4	38.3	12.4	43.1	16.1	21.5	27.1
		KD	53.4	25.9	40.4	13.4	43.4	16.9*	25.3	31.7
		SeqKD	52.0	25.6	38.9	14.0	42.4	15.9	26.1	32.9
		MiniLLM	**54.7**	**26.4**	**44.6***	**15.9**	**45.7**	**18.3***	**29.3***	**37.7***
OPT	13B	Teacher	70.3	29.2	56.1	18.4	58.0	17.8	30.4	36.1
	1.3B	SFT w/o KD	52.6	26.0	37.7	11.4	40.5	15.6	23.1	28.4
		KD	52.7	25.4	36.0	12.2	40.8	14.9	21.9	27.0
		SeqKD	51.0	26.1	36.6	12.7	42.6	16.6	21.4	28.2
		MiniLLM	**60.7**	**26.7**	**47.0**	**14.8**	**50.6**	**17.9***	**28.6**	**33.4**
	2.7B	SFT w/o KD	55.4	27.1	38.9	13.9	44.8	16.6	24.9	32.3
		KD	60.5	25.9	48.6	13.8	51.3	16.7	26.3	30.2
		SeqKD	57.6	27.5	40.5	13.3	44.5	16.5	25.3	32.3
		MiniLLM	**63.2**	**27.4**	**52.7**	**17.2**	**55.9**	**19.1***	**30.7***	**35.1**
	6.7B	SFT w/o KD	67.9	27.6	56.4	16.4	57.3	17.8	30.3	28.6
		KD	68.6	28.3	58.0	17.0	57.0	17.5	30.7*	26.7
		SeqKD	69.6	28.5	54.0	17.0	57.6	17.9*	30.4	28.2
		MiniLLM	**70.8***	**29.0**	**58.5***	**17.5**	**60.1***	**18.7***	**32.5***	**36.7***
LLaMA	13B	Teacher	79.0	29.7	75.5	23.4	65.1	19.4	35.8	38.5
	7B	SFT w/o KD	73.0	26.3	69.2	20.8	61.6	17.5	32.4	35.8
		KD	73.7	27.4	70.5	20.2	62.7	18.4	33.7	37.9
		SeqKD	73.6	27.5	71.5	20.8	62.6	18.1	33.7	37.6
		MiniLLM	**76.4**	**29.0**	**73.1**	**23.2**	**64.1**	**20.7***	**35.5**	**40.2***

Supplemental Results



Performance scales with size



Performance stays consistent with increased ground truth length

Ablation Studies Result

	Valid. R-L	Dolly R-L
MiniLLM	**27.4**	**24.6**
w/o Length Norm.	17.4	14.7
w/o Teacher-Mixed	22.3	20.4
w/o Single-Step	27.0	23.7

- Removing length normalization significantly degrades performance
- Teacher-mixed sampling shows strong upside
- Single-Step Decomposition had limited influence

Paper 3: LLM Pruning and Distillation in Practice:

The Minitron Approach, (NVIDIA - 2024)

Motivation:

Model Families from Scratch:

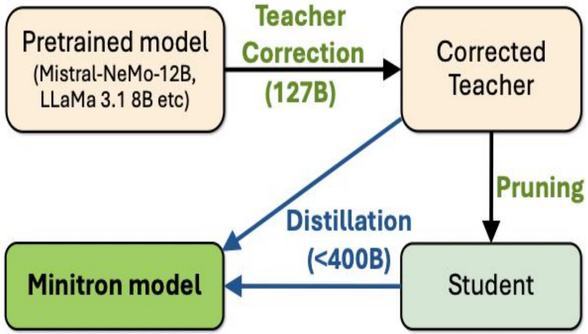
- LLM providers train entire model families to meet diverse deployment needs.
- This approach is **time-, data-, and resource-intensive.**

Challenges:

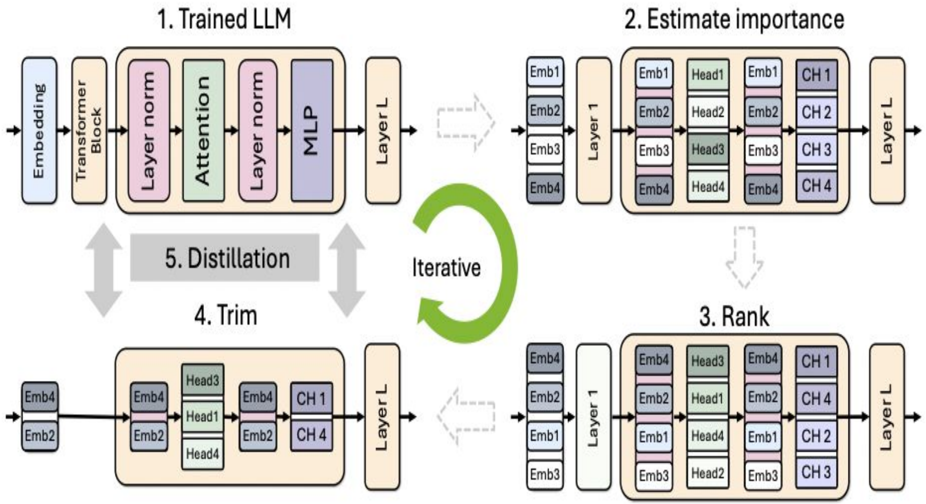
- Training multiple large models from scratch limits accessibility and practicality.
- Traditional approaches rely heavily on original training data.

Solution: Pruning & Distillation:

- Combine **weight pruning** and **knowledge distillation** to reduce costs.
- Only the largest model is trained from scratch.
- Smaller models are derived via pruning and accuracy recovery using distillation.
- **Teacher correction** to adapt the teacher for distillation on new datasets.



Methodology



1) Teacher Model Preparation:

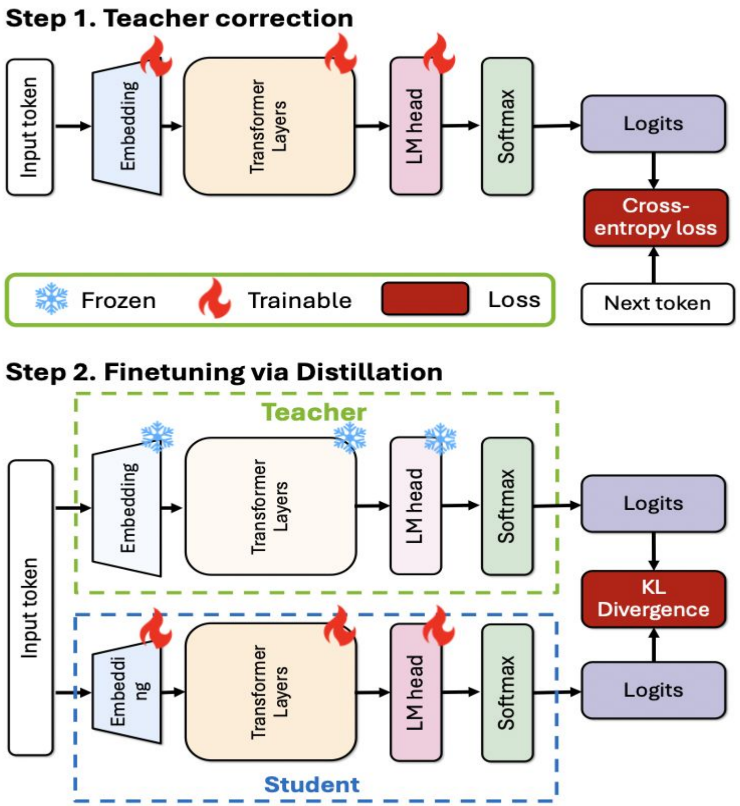
- Start with a large, pre-trained LLM (Llama 3.1 8B, Mistral NeMo 12B).
- Apply **teacher correction** by fine-tuning the model on the distillation dataset.

2) Pruning:

- **Importance Estimation:** Compute importance scores for layers, neurons, and attention heads using activation-based metrics.

- **Width Pruning:** Compress dimensions of hidden layers and MLPs while retaining attention heads.

- **Depth Pruning:** Reduce the number of layers by removing contiguous blocks based on importance.



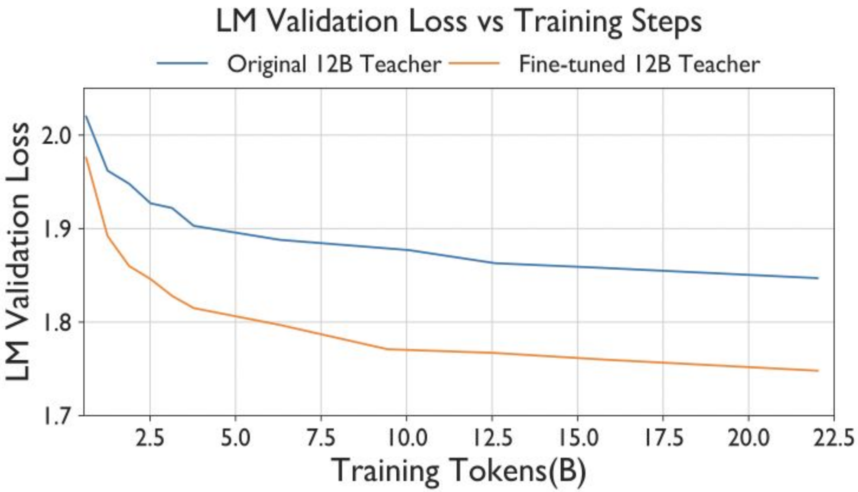
3) Knowledge Distillation:

- Transfer knowledge from the corrected teacher model to the smaller, pruned student model.
- Use **KL divergence** loss to ensure the student mimics the teacher's output distribution.

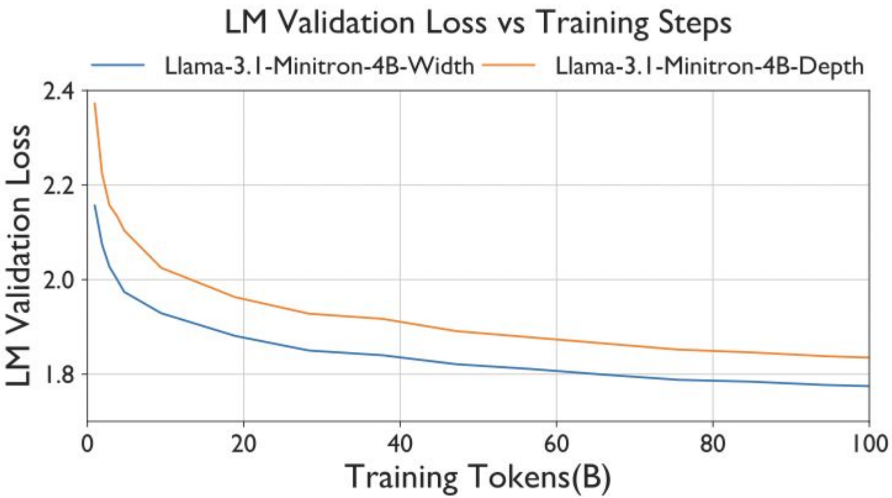
4) Instruction Tuning

- Applied to the Llama-3.1-Minitron-4B models using **NeMo-Aligner** to evaluate instruction-following capabilities

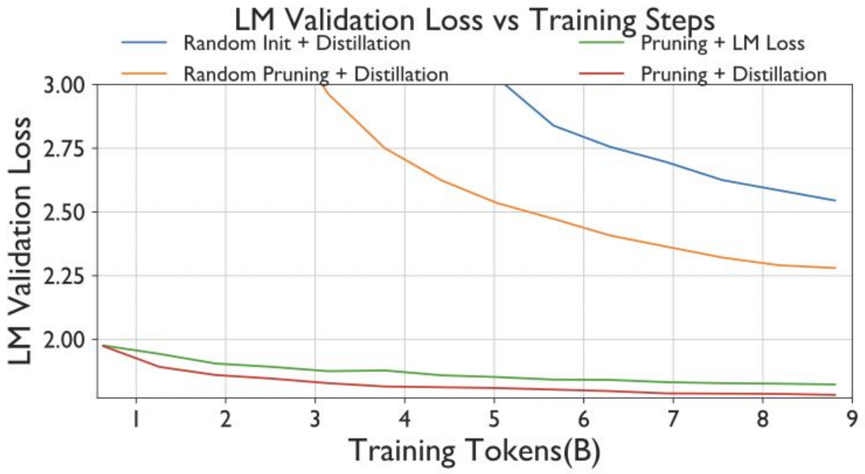
Experimental Results



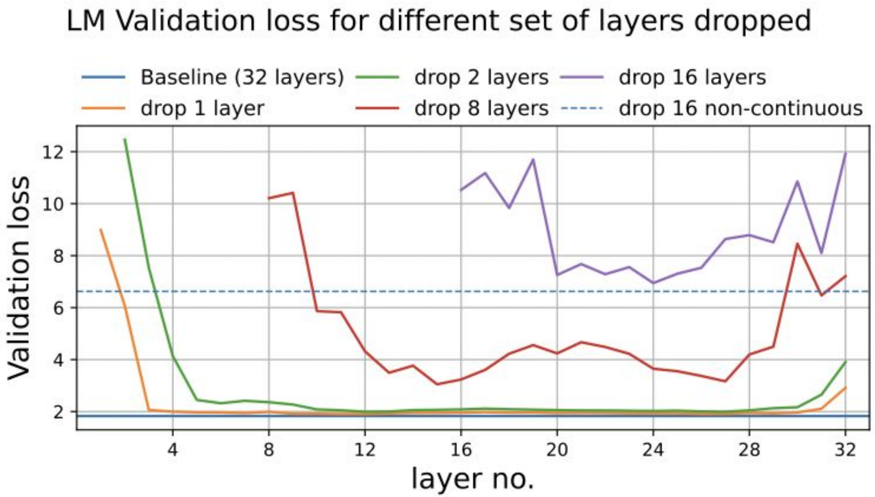
Training convergence plot for the compressed 8B student model. We compare supervision from the original teacher and the corrected teacher.



Convergence of width- & depth-pruned Llama 3.1 8B to 4B models. Width pruning outperforms depth pruning for a given parameter budget.



Training convergence plot for Mistral Nemo 12B compressed model. We compare (a) random initialization with distillation, (b) randomly pruned weights with distillation, (c) pruning with standard LM loss, and (d) our pipeline with pruning and distillation.



LM loss value on validation set after removing 1, 2, 8 or 16 contiguous layers with Llama 3.1 8B. For example, the purple line at layer no. 16 indicates the LM loss if we dropped the first 16 layers.

Benchmarks(shots)	Gemma2	Minitron	**Llama-3.1-Minitron**		Gemma	Mistral	Llama 3.1	**MN-Minitron**	Mistral NeMo	
	2B*	4B	**4B-Depth**	**4B-Width**	7B	7B	8B	**8B**	12B-Base	12B-FT
Total Params	2.6B	4.2B	4.5B	4.5B	8.5B	7.3B	8B	8.4B	12.2B	12.2B
Non-Emb. Params	2B	2.6B	3.7B	3.7B	7.7B	7B	7B	7.3B	10.9B	10.9B
Training Tokens	2T	**94B**	**94B**	**94B**	6T	8T	15T	**380B**	-	+0.1T
Winogrande(5)	70.9	**74.0**	72.1	73.5	78	78.5	77.3	**80.4**	82.2	82.7
Arc_challenge(25)	55.4	50.9	52.6	**55.6**	61	60.3	57.9	**64.4**	65.1	62.3
MMLU(5)	51.3	58.6	58.7	**60.5**	64	64.1	65.3	**69.5**	69.0	70.1
Hellaswag(10)	73.0	75.0	73.2	**76.1**	82	**83.2**	81.8	83.0	85.2	85.3
GSM8k(5)	23.9	24.1	16.8	**41.2**	50	37.0	48.6	**58.5**	56.4	55.7
Truthfulqa(0)	-	**42.9**	38.2	**42.9**	45	42.6	45.0	**47.6**	49.8	48.3
XLSum en(20%) (3)	-	**29.5**	27.2	28.7	17	4.8	30.0	**32.0**	33.4	31.9
MBPP(0)	29.0	28.2	30.7	**32.4**	39	38.8	42.3	**43.8**	42.6	47.9
HumanEval(n=20)(0)	20.1	**23.3**	-	-	32.0	28.7	24.8	**36.2**	23.8	23.8

Accuracy numbers for MN-Minitron-8B and Llama-3.1-Minitron-4B models. The two models are compared to similarly-sized SoTA open models on a variety of common language modeling benchmarks.

Benchmarks	Gemma	Phi-2	Gemma2	Qwen2	Minitron	**Llama-3.1-Minitron**	
	2B	2.7B	2B	1.5B	4B	**4B-Depth**	**4B-Width**
Total Params	2.5B	2.7B	2.6B	1.5B	4.2B	4.5B	4.5B
Non-Emb. Params	2B	2.5B	2B	1.3B	2.6B	3.5B	3.7B
Tokens	3T	1.4T	2T	7T	**94B**	**94B**	**94B**
IFEval	40.5	44.0	**64.5**	39.8	44.8	42.6	52.4
MT-Bench	5.2	4.3	**7.7**	5.2	5.6	5.6	6.3
ChatRAG*	33.3	37.6	37.5	32.8	41.1	40.1	**44.0**
BFCL	47.0	23.1	35.6	32.8	64.2	**66.8**	64.9

Accuracy numbers for the aligned Llama-3.1-Minitron models. The model is compared to similarly-sized SoTA open aligned models on a variety of benchmarks. Best in **bold**, second underlined.

Impact

- **Performance:**
  - The compressed models (4B and 8B) maintain or exceed the accuracy of their larger counterparts with up to **150× fewer training tokens**.
- **Practical Utility:**
  - Enables deployment of high-performing models in resource-constrained environments (e.g., edge devices).
- **Efficiency Gains:**
  - Up to **2.7× speedup** in runtime inference for pruned models.
  - Significant reduction in training tokens: MN-Minitron-8B (380B tokens vs. 15T for the teacher).

Summary

Distillation

The process of training a smaller "student" model to replicate the behavior of a larger "teacher" model by learning from its outputs rather than the original dataset.

Distilling Step by Step

Rationales as supervision for training small student models gives improved performance using only fraction of a dataset

MiniLLM

Optimizing Reverse KL-Divergence more effectively distills knowledge in LLMs and ensures diversity.

Minitron

Teacher correction & pruning strategies significantly enhance model performance, accuracy, and efficiency during distillation.

Thank You!